ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 6

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Compare Waterfall, Spiral, and Agile Scrum process models with lifecycle diagrams. [14 Marks]

[OR]

Q.1 (Alt) Explain Software Requirements Specification (SRS) IEEE standards and Functional vs Non-functional requirements. [14 Marks]

--- UNIT II ---

Q.2 Explain Software Cost Estimation using COCOMO-I and COCOMO-II models. Calculate effort. [14 Marks]

[OR]

Q.2 (Alt) Describe Risk Management: Risk Identification, Assessment, Projection, RMMM plan. [14 Marks]

--- UNIT III ---

Q.3 Explain Software Architectural Styles: MVC, Layered, Microservices. Explain Cohesion and Coupling. [14 Marks]

[OR]

Q.3 (Alt) Draw UML Use Case, Class, Sequence, and Activity diagrams for Online Banking System. [14 Marks]

--- UNIT IV ---

Q.4 Compare Black-Box vs White-Box testing. Calculate Cyclomatic Complexity using control flow graph. [14 Marks]

[OR]

Q.4 (Alt) Explain Unit, Integration, System, Acceptance, and Regression testing methodologies. [14 Marks]

--- UNIT V ---

Q.5 Explain Software Quality Assurance (SQA), ISO 9001, and CMMI maturity levels. [14 Marks]

[OR]

Q.5 (Alt) Describe DevOps pipeline, Continuous Integration/Continuous Deployment (CI/CD), and Git workflows. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****